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Electronic Components | Japan

Apr HDD/SSD Data: Continuing Strong Data Center Demand

Key Takeaways

- DC storage likely to stay segmented between SSDs, HDDs, and tape drives.
- TSR forecasts 2026 DC-use enterprise SSD shipment capacity at 476.1EB, up 81.3% YoY.
- TSR forecast 2026 DC-use NL HDD shipment capacity at 1,844EB, up 30.3% YoY.
- Non-enterprise SSD shipment volume for PCs, add-ons, and gaming declined YoY.
- TDK resumed HDD shipments to Seagate and Western Digital from Jan-Mar. We expect substantial volume growth from Apr-Jun.

No change to view for continued growth in shipment volume/capacity of both NL HDDs and enterprise SSDs for data center use:

We anticipate high production and per/unit capacity growth will continue for both NL HDDs and enterprise SSDs, and that overall shipment capacity will continue to rise. Techno Systems Research (TSR) forecasts further growth in total HDD shipment capacity, raising its 2026 estimate from 2,004.5EB (+23.1% YoY) last month to 2,058.2EB (+26.4% YoY) this month, compared to 1,627.9EB (+29.5% YoY) in 2025, and in total SSD shipment capacity, raising its 2026 estimate from 668.9EB (+34.3% YoY) last month to 675.30EB (+35.6% YoY) this month, compared to 498.0EB in 2025 (+38.0% YoY). With higher-capacity products accounting for a rising share of the mix in both NL HDDs and enterprise SSDs, we see potential for capacity growth to significantly outstrip shipment unit growth. Meanwhile, shipment volumes of non-enterprise SSDs for PCs, add-ons, and gaming declined YoY, partly due to rising NAND flash prices.

TDK resumed shipments of HDD heads to Seagate and Western Digital in Jan-Mar quarter:

According to TSR data, Jan-Mar shipments of heads to Seagate (200,000 units) accounted for just 0.4% of TDK's total shipment volume (52.8mn units) and 0.1% of Seagate head usage (165.6mn units) while shipments to WDC (700,000 units) accounted for only 1.3% of TDK's shipments and 0.4% of WDC's usage (184.4mn units). However, TSR expects that TDK's shipments of heads to Seagate and WDC will rise sharply from Apr-Jun. TDK has indicated that it expects 40% YoY growth in F3/27 HDD head shipment volume, suggesting a substantial increase from 198.7mn units in F3/26 to 278.18mn units in F3/27.

Apr DC NL HDD output +14.3% YoY, -0.6% MoM: Apr NL HDD production was 6.80mn units (5.95mn a year ago, 6.84mn a month ago). NL output fell from 81.45mn units in 2021 (+18.0% YoY) to 71.30mn in 2022 (-12.5%) and 44.51mn in 2023 (-37.6%), then rose to 63.12mn in 2024 (+41.8%) and to 74.77mn in 2025 (+18.5%). TSR forecasts 83.97mn units for 2026 (+12.3% YoY), vs last month's forecast of 84.22mn (+12.6% YoY), with Jan-Mar production at 20.77mn units (+19.9% YoY, +5.0% QoQ) and Apr-Jun at 20.85mn (+13.5% YoY, +0.4% QoQ).

MORGAN STANLEY MUFG SECURITIES CO., LTD. +

Shoji Sato

Equity Analyst

Shoji.Sato@morganstanleymufg.com

+81 3 6836-8404

Sota Harashima

Equity Analyst

Sota.Harashima@morganstanleymufg.com

+81 3 6836-8897



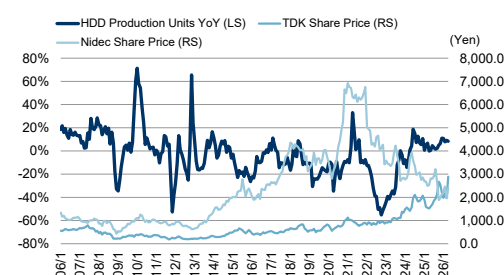
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Industry View

In-Line

Exhibit 1 : HDD Production Volume YoY and Share Prices of TDK & Nidec



Source: TSR, Morgan Stanley Research

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Apr DC-use enterprise SSD shipments (capacity) +116.7% YoY, +2.5% MoM: Apr enterprise SSD shipments came to 37.70EB (17.40EB a year ago, 36.80EB a month ago). Full CY: 77.47EB in 2021, 109.83EB (+41.8% YoY) in 2022, 102.66EB (–6.5%) in 2023, 165.95EB (+61.6%) in 2024, and 262.5EB (+58.2% YoY) in 2025. TSR forecasts total enterprise SSD shipment capacity of 476.1EB in 2026 (+81.30% YoY), up from last month's forecast of 425.3EB (+62.0%).

Apr HDD output +8.3% YoY, -2.4% MoM: Apr HDD production was 11.18mn units (10.32mn a year ago, 11.45mn a month ago). HDD output declined from 261.70mn units in 2021 (–0.1% YoY) to 168.70mn in 2022 (–35.5%), then to 120.51mn in 2023 (–28.6%), but rose to 124.77mn in 2024 (+3.5%) and to 129.54mn in 2025 (+3.8%). TSR forecasts 135.04mn units for 2026 (+4.2% YoY), up from last month's forecast of 134.86mn (+4.1% YoY), estimating Jan–Mar at 33.89mn units (+12.9% YoY, +0.1% QoQ) and Apr–Jun at 33.81mn units (+6.7% YoY, –0.2% QoQ), vs last month's forecast of 33.75mn (+6.5% YoY, –1.2% QoQ). Our forecasts shown in Exhibit 4 onward reflect shipments and differ from TSR's production estimates.

Apr SSD shipments +3.3% YoY, -1.5% MoM: Apr SSD shipments were 29.23mn units (28.30mn a year ago, 29.67mn a month ago). SSD shipments fell from 383.02mn units in 2021 (+21.4% YoY) to 348.30mn in 2022 (–9.1%), then to 319.80mn in 2023 (–8.2%), but rose to 348.20mn in 2024 (+8.0%) and to 375.01mn in 2025. TSR forecasts shipments of 371.56mn units for 2026 (–0.9% YoY) and sees Jan–Mar at 89.20mn units (+2.6% YoY, –11.4% QoQ), versus Apr–Jun at 90.38mn units (+0.6% YoY, +1.3% QoQ), down from last month's forecast of 90.08mn (+0.3% YoY, +1.0% QoQ).

Apr DC-use enterprise SSD shipments +42.0% YoY, -3.7% MoM: Apr enterprise SSD shipments were 5.68mn units (4.00mn a year ago, 5.90mn a month ago). Enterprise SSD shipments rose from 36.10mn units in 2021 (+6.8% YoY) to 47.30mn in 2022 (+31.0%), then fell to 40.10mn in 2023 (–15.2%), but rose to 50.00mn in 2024 (+24.7%) and to 56.51mn in 2025 (+13.1%). TSR forecasts 71.53mn units for 2026 (+26.6% YoY), down from last month's forecast of 72.31mn (+28.0% YoY), estimating Jan–Mar at 17.20mn units (+41.0% YoY, +2.9% QoQ), and Apr–Jun at 17.63mn units (+41.0% YoY, +2.5% QoQ), unchanged from last month's forecast.

Exhibit 2: HDD Production Volume by Company and Size

(k units)		25/4	25/5	25/6	25/7	25/8	25/9	25/10	25/11	25/12	26/1	26/2	26/3	26/4	YoY	MoM
Seagate	Enterprise	150	150	150	150	150	150	150	100	160	110	110	160	140	-6.7%	-12.5%
	NL	2,400	2,500	2,600	2,600	2,650	2,650	2,700	2,800	2,900	2,850	2,870	2,700	2,800	16.7%	3.7%
	3.5"ATA	1,050	1,050	900	1,000	1,000	1,000	1,000	900	1,100	1,000	950	900	950	-9.5%	5.6%
	2.5"Mobile	600	600	700	700	650	650	650	650	650	600	550	600	550	-8.3%	-8.3%
	Total	4,400	4,300	4,650	4,450	4,450	4,450	4,500	4,450	4,810	4,560	4,480	4,360	4,440	5.7%	1.8%
WDC	Enterprise															
	NL	2,800	2,900	2,900	2,900	2,950	2,950	2,960	3,160	3,080	3,150	3,200	3,100	3,200	14.3%	3.2%
	3.5"ATA	1,000	1,000	1,100	1,200	1,150	1,150	1,100	1,050	1,100	1,150	1,100	1,100	1,050	5.0%	-4.5%
	2.5"Mobile	600	600	650	650	700	650	600	600	600	580	580	650	600	0.0%	-7.7%
	Total	4,400	4,500	4,650	4,750	4,800	4,750	4,660	4,810	4,780	4,880	4,880	4,850	4,850	10.2%	0.0%
Toshiba	Enterprise	130	140	130	130	110	100	140	160	140	120	90	100	60	-53.8%	-40.0%
	NL	750	760	760	720	770	770	920	870	730	880	980	1,040	800	6.7%	-23.1%
	3.5"ATA	230	220	300	240	210	350	240	415	330	320	270	500	330	43.5%	-34.0%
	2.5"Mobile	610	620	630	660	660	640	680	660	540	500	480	600	700	14.8%	16.7%
	Total	1,720	1,740	1,820	1,750	1,750	1,860	1,980	2,105	1,740	1,820	1,820	2,240	1,890	9.9%	-15.6%
Total	Enterprise	280	290	280	280	260	250	290	260	300	230	200	260	200	-28.6%	-23.1%
	NL	5,950	6,160	6,260	6,220	6,370	6,370	6,580	6,830	6,710	6,880	7,050	6,840	6,800	14.3%	-0.6%
	(Air)	1,360	1,310	1,190	1,170	1,245	1,295	1,270	1,280	1,280	1,220	1,210	1,120	1,070	-21.3%	-4.5%
	(He)	4,590	4,850	5,070	5,050	5,125	5,075	5,310	5,550	5,430	5,660	5,840	5,720	5,730	24.8%	0.2%
	3.5"ATA	2,280	2,270	2,300	2,440	2,360	2,500	2,340	2,365	2,530	2,470	2,320	2,500	2,330	2.2%	-6.8%
	2.5"Mobile	1,810	1,820	1,980	2,010	2,010	1,940	1,930	1,910	1,790	1,680	1,610	1,850	1,850	2.2%	0.0%
	Total	10,320	10,540	10,820	10,950	11,000	11,060	11,140	11,365	11,330	11,260	11,180	11,450	11,180	8.3%	-2.4%
YoY % Change	Enterprise	-37.8%	-9.4%	-12.5%	-15.2%	-27.8%	-21.9%	-17.1%	-27.8%	-9.1%	-23.3%	-42.9%	-13.3%	-28.6%		
	NL	21.4%	17.3%	19.7%	13.7%	11.8%	11.2%	15.4%	20.2%	14.5%	22.0%	20.5%	17.3%	14.3%		
	(Air)	-2.2%	-5.8%	-11.2%	-24.5%	-14.1%	-15.6%	-17.8%	-16.1%	-15.0%	-19.2%	-18.2%	-11.1%	-21.3%		
	(He)	30.8%	25.6%	30.3%	28.8%	20.6%	21.0%	27.8%	33.6%	24.7%	37.0%	33.6%	25.2%	24.8%		
	3.5"ATA	-17.7%	-17.5%	-14.2%	-11.9%	-9.9%	-9.1%	-4.9%	-8.0%	15.5%	5.1%	0.9%	1.2%	2.2%		
	2.5"	-18.1%	-15.7%	-6.6%	-7.8%	-6.5%	-4.0%	-8.1%	-8.2%	-2.2%	-10.2%	-13.4%	-2.1%	2.2%		
	Total	-0.1%	0.6%	4.5%	2.8%	1.6%	2.2%	5.0%	6.3%	11.0%	10.8%	7.9%	9.2%	8.3%		
MoM % Change	Enterprise	-6.7%	3.6%	-3.4%	0.0%	-7.1%	-3.8%	16.0%	-10.3%	15.4%	-23.3%	-13.0%	30.0%	-23.1%		
	NL	2.1%	3.5%	1.6%	-0.6%	2.4%	0.0%	3.3%	3.8%	-1.8%	2.5%	2.5%	-3.0%	-0.6%		
	(Air)	7.9%	-3.7%	-9.2%	-1.7%	6.4%	4.0%	0.8%	0.0%	-4.7%	-0.8%	-7.4%	-4.5%			
	(He)	0.4%	5.7%	4.5%	-0.4%	1.5%	-1.0%	4.6%	4.5%	-2.2%	4.2%	3.2%	-2.1%	0.2%		
	3.5"ATA	-7.7%	-0.4%	1.3%	6.1%	-3.3%	5.9%	-6.4%	1.1%	7.0%	-2.4%	-6.1%	7.8%	-6.8%		
	2.5"	-4.2%	0.6%	8.8%	1.5%	0.0%	-3.5%	-0.5%	-1.0%	-6.3%	-6.1%	-4.2%	14.9%	0.0%		
	Total	-1.6%	2.1%	2.7%	1.2%	0.5%	0.5%	0.7%	2.0%	-0.3%	-0.6%	-0.7%	2.4%	-2.4%		

Source: TSR, Morgan Stanley Research

Exhibit 3: SSD Shipments/Capacity by Application

(k units)		25/4	25/5	25/6	25/7	25/8	25/9	25/10	25/11	25/12	26/1	26/2	26/3	26/4	YoY	MoM
SSD shipment	Enterprise	4,000	4,000	4,500	4,870	5,010	5,220	5,500	5,610	5,600	5,630	5,670	5,900	5,680	42.0%	-3.7%
	PC	17,500	18,500	19,500	19,000	20,000	20,500	21,000	21,500	19,000	18,000	17,000	17,500	17,430	-0.4%	-0.4%
	Add-on	4,000	4,100	4,200	4,200	4,300	4,500	4,500	4,600	4,400	4,000	3,700	3,500	3,560	-11.0%	1.7%
	Game	2,000	2,400	2,600	2,000	2,500	3,000	2,500	2,000	2,000	1,800	1,800	1,800	1,600	-20.0%	-11.1%
	Others	800	800	900	800	800	900	800	800	900	950	980	970	960	20.0%	-1.0%
	Total	28,300	29,800	31,700	30,870	32,610	34,120	34,300	34,510	31,900	30,380	29,150	29,670	29,230	3.3%	-1.5%
YoY % Change	Enterprise	14.3%	-4.8%	-6.3%	13.3%	11.3%	8.7%	thly	19.4%	24.4%	34.0%	35.0%	55.3%	42.0%		
	PC	-2.8%	1.1%	5.4%	4.4%	8.1%	9.6%	10.5%	16.2%	9.2%	5.9%	-5.6%	-7.9%	-0.4%		
	Add-on	14.3%	10.8%	5.0%	13.5%	13.2%	15.4%	12.5%	12.2%	2.3%	2.6%	-7.5%	-14.6%	-11.0%		
	Game	33.3%	20.0%	30.0%	0.0%	25.0%	50.0%	25.0%	0.0%	66.7%	-5.3%	-10.0%	-25.0%	-20.0%		
	Others	0.0%	14.3%	12.5%	14.3%	0.0%	12.5%	0.0%	0.0%	12.5%	18.8%	22.5%	21.3%	20.0%		
	Total	3.7%	3.1%	5.3%	6.8%	10.2%	13.0%	11.4%	14.7%	13.1%	9.3%	0.5%	-1.4%	3.3%		
MoM % Change	Enterprise	5.3%	0.0%	12.5%	8.2%	2.9%	4.2%	5.4%	2.0%	-0.2%	0.5%	0.7%	4.1%	-3.7%		
	PC	-7.9%	5.7%	5.4%	-2.6%	5.3%	2.5%	2.4%	2.4%	-11.6%	-5.3%	-5.6%	2.9%	-0.4%		
	Add-on	-2.4%	2.5%	2.4%	0.0%	2.4%	4.7%	0.0%	2.2%	-4.3%	-9.1%	-7.5%	-5.4%	1.7%		
	Game	-16.7%	20.0%	8.3%	-23.1%	25.0%	20.0%	-16.7%	-20.0%	0.0%	-10.0%	0.0%	0.0%	-11.1%		
	Others	0.0%	0.0%	12.5%	-11.1%	0.0%	12.5%	-11.1%	0.0%	12.5%	5.6%	3.2%	-1.0%	-1.0%		
	Total	-6.0%	5.3%	6.4%	-2.6%	5.6%	4.6%	0.5%	0.6%	-7.6%	-4.8%	-4.0%	1.8%	-1.5%		
SSD capacity (EB)	Enterprise	17.40	17.32	20.64	22.40	23.51	24.60	28.02	28.88	30.63	32.91	35.03	36.80	37.70	116.7%	2.5%
	PC	11.06	11.71	12.85	12.85	13.62	14.06	14.55	14.94	13.44	12.93	12.22	12.61	12.60	13.9%	-0.1%
	Add-on	3.43	3.55	3.93	3.86	4.16	4.41	4.55	4.84	4.54	4.37	3.96	3.52	3.73	8.7%	6.0%
	Game	2.20	2.80	3.20	2.50	3.00	3.50	3.00	2.40	2.30	2.20	2.20	2.20	1.90	-13.6%	-13.6%
	Others	0.11	0.11	0.14	0.13	0.13	0.16	0.14	0.15	0.17	0.19	0.20	0.20	0.21	94.5%	5.3%
	Total	34.20	35.49	40.76	41.74	44.41	46.72	50.27	51.20	51.07	52.69	53.60	55.33	56.15	64.2%	1.5%

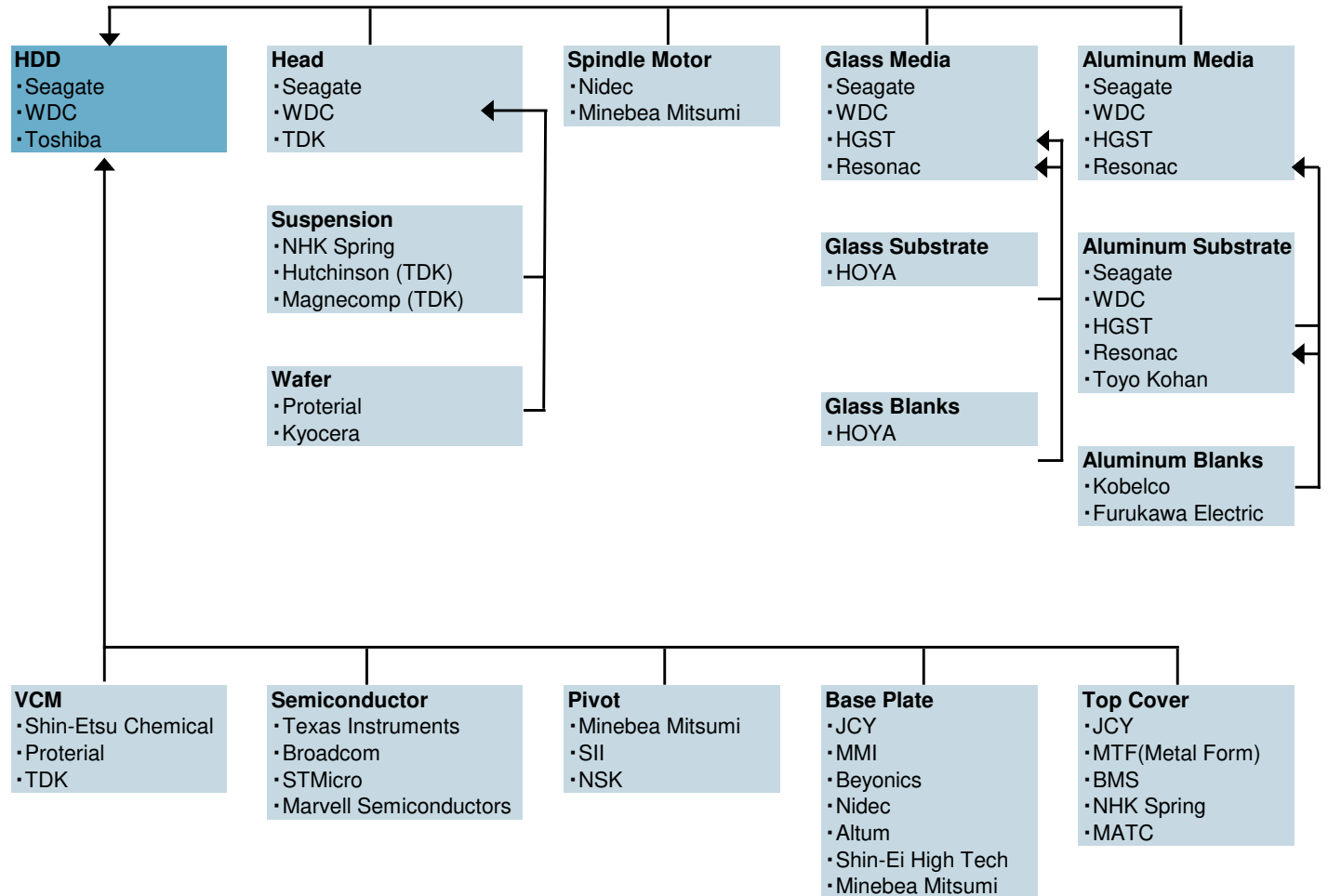
Source: TSR, Morgan Stanley Research

Exhibit 4: HDD Shipment Volume / Shipment Capacity

Note: e = Morgan Stanley Research estimates
Source: TSR, Morgan Stanley Research

HDD supply chain

Exhibit 5: HDD Supply Chain



Source: TSR, Morgan Stanley Research

Exhibit 6: Market Shares of HDD Parts Suppliers (as of Jan-Mar 2026)

HDD Head		Suppliers			HDD Motor		Suppliers		HD Media		Suppliers			
		Seagate	WDC	TDK			Nidec	Minebea			Seagate	WDC	HGST	Resonac
Customers	Seagate	99.9%		0.1%	Customers	Seagate	100%		Customers	Seagate	100%			
	Enterprise	100%				Enterprise	100%			Enterprise	100%			
	NL	99.9%		0.1%		NL	100%			NL	100%			
	3.5" ATA	100%				3.5" ATA	100%			3.5" ATA	100%			
	2.5"	100%				2.5"	100%			2.5"	100%			
	Samsung					Samsung				Samsung				
	3.5" ATA					3.5" ATA				3.5" ATA				
	2.5"					2.5"				2.5"				
	WDC		100%			WDC	100%			WDC		100%		
	Enterprise					Enterprise				Enterprise				
	NL					NL				NL				
	3.5" ATA		100%			3.5" ATA	100%			3.5" ATA		100%		
	2.5"		100%			2.5"	100%			2.5"		100%		
	HGST		99.6%	0.4%		HGST	100%			HGST			93%	7%
	Enterprise					Enterprise				Enterprise				
	NL		99.6%	0.4%		NL	100%			NL			93%	7%
	3.5" ATA					3.5" ATA				3.5" ATA				
	2.5"					2.5"				2.5"				
Customers	Toshiba			100%	Customers	Toshiba	100%		Customers	Toshiba				100%
	Enterprise			100%		Enterprise	100%			Enterprise				100%
	NL			100%		NL	100%			NL				100%
	3.5" ATA			100%		3.5" ATA	100%			3.5" ATA				100%
	2.5"			100%		2.5"	100%			2.5"				100%
	1.8"					1.8"				1.8"				
	Total	41%	46%	13%		Total	60%	40%		Total	42%	4%	40%	15%
	Enterprise	65%		35%		Enterprise	49%	51%		Enterprise	63%			37%
	NL	42%	47%	12%		NL	58%	42%		NL	42%		44%	14%
	3.5" ATA	37%	47%	16%		3.5" ATA	64%	36%		3.5" ATA	38%	50%		12%
	2.5"	33%	38%	29%		2.5"	66%	34%		2.5"	35%	40%		25%

Source: TSR, Morgan Stanley Research

Exhibit 7: HDD Head Shipment Volume

	CY23				CY24				CY25				CY26				CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q												
HDD Head Shipment																												
Total	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9				1,931.6	1,756.1	1,702.8	1,714.4	1,715.4	1,496.2	1,517.9	1,747.9	1,349.0	919.9	1,219.3	1,454.1
TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8				471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8
Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4				634.0	616.2	617.1	636.6	683.5	594.1	652.2	755.2	579.1	399.5	492.9	589.4
Western Digital	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7				826.0	763.9	710.7	715.9	701.7	602.5	611.3	654.5	539.1	379.1	543.0	670.9
Market Share																												
TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%				24%	21%	22%	21%	19%	20%	17%	19%	17%	15%	15%	13%
Seagate	45%	45%	40%	43%	39%	39%	41%	42%	39%	40%	41%	42%	41%				33%	35%	36%	37%	40%	40%	43%	43%	43%	43%	40%	41%
Western Digital	40%	42%	44%	40%	44%	45%	44%	45%	47%	47%	46%	46%	46%				43%	43%	42%	42%	41%	40%	40%	37%	40%	41%	45%	46%
YoY % Change																												
TDK	-38%	-47%	-33%	9%	0%	46%	51%	40%	26%	17%	15%	20%	22%				0%	-9%	-3%	1%	0%	-13%	1%	15%	-23%	-32%	33%	19%
Seagate	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%				-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%
Western Digital	-40%	-47%	-32%	26%	-13%	26%	53%	38%	24%	21%	17%	17%	23%				1%	-3%	0%	3%	7%	-13%	10%	16%	-23%	-31%	23%	20%
Western Digital	-33%	-44%	-31%	2%	12%	58%	53%	55%	33%	22%	19%	22%	19%				2%	-8%	-7%	1%	-2%	-14%	1%	7%	-18%	-30%	43%	24%
HDD Head/HDD	7.84	6.49	7.59	8.22	8.83	9.80	10.20	10.47	11.50	11.47	11.95	12.04	12.18				3.42	3.75	4.02	4.25	4.57	4.71	5.84	6.74	7.83	7.52	9.84	11.75

Note: e = Morgan Stanley Research estimates

Source: TSR, Morgan Stanley Research

Exhibit 8: Trends in In-house Production of HDD Heads/Media

(mn units)	CY23				CY24				CY25				CY26				CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q															
HDD Head																												
Seagate																												
Total demand	123.1	95.8	87.9	102.0	103.4	116.2	132.9	140.7	128.3	140.8	155.1	165.2	165.6				781.6	717.2	664.2	677.2	693.7	650.6	695.3	800.7	607.2	408.8	493.2	589.4
Volume of in-house production	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4				634.0	616.2	617.1	653.7	646.2	600.3	652.2	755.2	579.1	399.5	492.9	589.4
Shipment volume from TDK	4.3	3.7	1.0	0.3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.2				147.6	101.0	47.1	23.5	47.5	50.3	43.1	45.5	28.1	9.3	0.3	0.0
Ratio of in-house production	97%	96%	99%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.9%				81%	86%	93%	97%	93%	92%	94%	94%	95%	98%	100%	100%
WD																												
Total demand	106.7	86.4	95.4	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	184.4				882.8	797.4	738.8	749.2	676.2	612.1	611.3	659.3	546.7	384.2	543.0	670.9
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2				526.2	460.8	386.4	395.0	314.4	228.2	216.5	195.5	95.1	73.2	59.7	59.4
ex-HGST	86.0	67.6	78.2	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	169.2				356.6	336.6	352.4	354.2	361.8	383.9	394.8	463.8	451.6	311.0	483.3	611.5
Volume of in-house production	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7				826.0	763.9	710.7	717.0	654.2	611.2	611.3	654.5	539.1	379.1	543.0	670.9
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2				482.5	443.3	370.2	385.3	309.4	227.3	216.5	195.5	95.1	73.2	59.7	59.4
ex-HGST	83.1	65.6	78.0	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	168.5				343.5	320.6	340.5	331.7	344.8	383.9	394.8	459.0	444.0	305.9	483.3	611.5
Shipment volume from TDK	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7				56.8	33.5	28.1	32.2	22.0	0.9	0.0	4.8	7.6	5.1	0.0	0.0
Ratio of in-house production	97%	98%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.6%				94%	96%	96%	96%	97%	100%	100%	99%	99%	99%	100%	100%
Toshiba																												
Total demand	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9				267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shipment volume from TDK	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9				267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8
Ratio of in-house production	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Industry Total																												
Total demand	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9				1,931.6	1,757.1	1,702.8	1,720.6	1,622.1	1,516.8	1,517.9	1,747.9	1,349.0	1,185.1	1,219.3	1,454.1
Shipment volume from TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8				471.6	377.0	375.0	349.9	321.7	305.3	254.4	338.2	230.8	198.1	183.4	193.8
Market share of TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%				24%	21%	22%	20%	20%	17%	19%	17%	17%	15%	15%	13%
HDD Media																												
Seagate																												
Total demand	64.1	48.7	44.8	51.8	53.2	59.8	67.0	71.0	67.7	73.9	77.6	77.6	77.6				395.5	359.0	330.6	330.3	363.6	327.1	351.8	404.7	308.7	209.4	251.0	296.8
Volume of in-house production	59.6	45.7	42.8	51.8	53.2	59.8	67.0	71.0	67.7	73.9	77.6	77.6	77.6				283.1	258.7	280.6	301.6	325.1	281.6	306.1	363.8	279.5	199.9	251.0	296.8
Shipment volume from Resonac	4.5	3.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				61.4	48.0	16.9	0.0	0.0	0.0	9.5	29.3	29.2	9.5	0.0	0.0
Ratio of in-house production	93%	94%	96%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%				72%	72%	85%	91%	89%	86%	87%	90%	91%	95%	100%	100%
WD																												
Total demand	55.2	44.0	48.5	48.8	60.1	69.0	73.2	74.8	80.0	84.8	87.4	87.4	87.4				442.9	398.7	369.9	374.8	365.8	305.8	309.4	333.9	276.3	196.5	277.1	339.6
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.9	7.9				263.1	230.4	193.2	196.2	171.4	125.0	111.9	102.0	50.5	39.9	32.7	31.2
ex-HGST	44.1	33.8	39.1	39.6	51.7	60.8	65.0	66.9	72.5	76.9	79.5	79.5	79.5				179.8	168.3	176.7	178.6	194.4	180.8	197.5	231.9	225.8	156.6	244.4	308.4
Volume of in-house production	49.7	40.0	45.5	45.8	57.1	64.2	68.2	69.8	75.0	79.8	82.4	82.4	82.4				399.9	368.0	343.9	337.0	339.7	309.9	309.4	333.3	259.2	181.0	259.3	319.6
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.9	7.9				233.1	211.9	177.2	182.2	163.4	123.0	111.9	102.0	50.5	39.9	32.7	31.2
ex-HGST	38.6	29.8	36.1	36.6	48.7	56.0	60.0	61.9	67.5	71.9	74.5	74.5	74.5				166.8	156.1	166.7	148.5	176.3	177.9	197.5	230.3	208.7	141.1	226.6	288.4
Shipment volume from Resonac	5.5	4.0	3.0	3.0	4.8	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0				43.0	30.7	26.0	44.1	26.1	3.2	0.0	1.6	17.1	15.5	17.8	20.0
Ratio of in-house production	90%	91%	94%	94%	95%	93%	93%	93%	94%	94%	94%	94%	94%				90%	92%	93%	88%	93%	98%	100%	100%	94%	92%	94%	94%
Toshiba																												
Total demand	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	26.0	26.0				135.6	122.0	149.9	152.1	133.0	124.3	107.1	145.8	105.3	66.7	96.4	105.2
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shipment volume from Resonac	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	26.0	26.0				135.6	122.0	149.9	152.1	133.0	124.3	107.1	146.1	105.3	66.7	96.4	105.2
Ratio of in-house production	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Industry Total																												
Total demand	135.8	103.9	112.3	120.6	136.3	151.2	166.2	170.8	176.9	182.7	191.0	191.0	191.0				974.0	879.7	850.4	856.2	862.4	757.5	768.3	884.4	690.3	472.6	624.5	741.6
Shipment volume from Resonac	26.5	18.2	24.0	23.0	26.0	27.2	31.0	30.0	34.2	29.0	31.0	31.0	31.0				240.0	210.7	192.8	196.2	159.1	127.2	116.6	177.0	151.6	91.7	114.2	125.2
Market share of Resonac	20%	18%	21%	19%	19%	18%	19%	18%	19%	16%	16%	16%	16%				25%	24%	23%	23%	18%	17%	15%	20%	22%	19%	18%	17%
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				51.0	42.3	33.1	28.7	38.5	45.5	36.2	11.6	0.0	0.0	0.0	0.0
Market share of Fuji Elec	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				5%	5%	4%	3%	4%	6%	5%	1%	0%	0%	0%	0%
HDD Media/HDD	4.06	3.32	3.92	4.17	4.58	4.99	5.18	5.38	6.16	6.01	6.04	5.78	5.79				1.73	1.88	2.01	2.13	2.30	2.38	2.96	3.41	4.01	3.87	5.04	5.99

Exhibit 9: HDD Motor Shipment Volume

	CY23				CY24				CY25				CY26															
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
HDD Motor Shipment																												
Total	34.2	27.2	31.6	29.5	30.5	32.7	33.2	32.6	30.8	31.7	32.5	34.3	34.0				575.2	476.5	431.7	407.3	381.6	307.7	271.8	248.8	146.1	122.5	129.0	129.3
Nidec	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4				408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0	77.8
Minebea Mitsumi	15.0	14.0	12.5	12.1	11.9	13.8	13.9	13.4	12.0	12.9	12.5	14.1	13.6				70.9	67.7	69.1	65.4	63.8	57.5	60.9	115.1	64.0	53.6	53.0	51.5
Market Share																												
Nidec	56%	49%	60%	59%	61%	58%	58%	59%	61%	59%	62%	59%	60%				71%	80%	84%	84%	83%	81%	78%	54%	56%	56%	59%	60%
Minebea Mitsumi	44%	51%	40%	41%	39%	42%	42%	41%	39%	41%	38%	41%	40%				12%	14%	16%	16%	17%	19%	22%	46%	44%	44%	41%	40%
YoY % Change																												
Nidec	-32%	-27%	2%	8%	-11%	20%	5%	11%	1%	-3%	-2%	5%	10%				3%	-17%	-9%	-6%	-6%	-19%	-12%	-8%	-41%	-16%	5%	0%
Nidec	-25%	-30%	3%	-9%	-3%	43%	1%	10%	1%	-1%	4%	5%	9%				4%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%	2%
Minebea Mitsumi	-40%	-25%	1%	49%	-21%	-1%	11%	11%	1%	-7%	-10%	5%	13%				-4%	-5%	2%	-5%	-2%	-10%	6%	89%	-44%	-16%	-1%	-3%
Sequential % Change																												
Nidec	25%	-20%	16%	-7%	3%	7%	2%	-2%	-6%	3%	3%	6%	-1%				-	-	-	-	-	-	-	-	-	-	-	-
Nidec	0%	-31%	45%	-9%	7%	2%	2%	-1%	-2%	0%	6%	1%	1%				-	-	-	-	-	-	-	-	-	-	-	-
Minebea Mitsumi	85%	-7%	-11%	-3%	-2%	16%	1%	-4%	-10%	8%	-3%	13%	-4%				-	-	-	-	-	-	-	-	-	-	-	-
HDD Motor/HDD	1.02	0.87	1.10	1.02	1.03	1.08	1.03	1.03	1.07	1.04	1.03	1.04	1.03				1.02	1.02	1.02	1.01	1.02	0.97	1.05	0.96	0.85	1.00	1.04	1.04

Source: TSR, Morgan Stanley Research

Disclosure Section

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(as of April 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Electronic Components

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/29/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	O (03/17/2026)	¥2,171
Hamamatsu Photonics (6965.T)	U (09/17/2025)	¥2,863
Ibiden (4062.T)	U (02/04/2026)	¥23,000
Kyocera (6971.T)	E (06/25/2020)	¥3,483
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,552
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥4,562
Murata Manufacturing (6981.T)	O (11/26/2025)	¥9,625
Nidec (6594.T)	NR (09/05/2025)	¥2,792
Niterra (5334.T)	O (01/17/2024)	¥10,165
Taiyo Yuden (6976.T)	E (10/19/2023)	¥14,815
TDK (6762.T)	O (08/02/2022)	¥4,108
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥768
Daishinku (6962.T)	E (03/07/2024)	¥1,019
Hirose Electric (6806.T)	O (07/10/2024)	¥28,225
IRISO Electronics (6908.T)	E (08/02/2022)	¥3,220
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,527
KOA (6999.T)	U (11/04/2025)	¥2,770
Meiko Electronics (6787.T)	E (04/03/2026)	¥38,600
Nichicon (6996.T)	E (11/10/2021)	¥3,770
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥3,195
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥5,080

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* Historical prices are not split adjusted.